

Are You Getting into Hot Water?

*We've come a long way in making hot water more efficiently.
The next place to cut waste is in the distribution systems.*

by **Robert Wendt**
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Have you ever wanted to take a quick, hot shower and had to wait and wait for hot water to arrive? Have you noticed that your dishwasher or clothes washer doesn't seem to be getting the hot water it needs to work effectively? And have you ever wondered what all that "cool" water down the drain is costing you, and what you can do to fix the problem?

The problems described above are limited to just a few older homes, where years of exposure to hard water have plugged the hot water pipes, slowing the flow to a trickle—right? *Wrong!* Energy and water waste in hot water distribution systems exists both in older and in brand-new homes. In fact, the newest homes built in the United States can have the greatest problems. As houses have grown in size, the distribution systems have increased in length. Modern home floor plans disperse the devices that consume hot water (lavatories, showers, clothes washers, dishwashers) throughout the house. There are more bathrooms per house than ever before, and fewer people to use them. Current plumbing codes developed in the 1940s do not reflect the household demographic changes and water-conserving technologies of today and typically result in oversized pipes that hold and waste more hot water.

The technological responses to these problems haven't always improved the situation. Larger, upscale homes are frequently built with a continuously operating hot water recirculation loop. Even though it is usually insulated, this loop serves as a large "radiator" to dissipate the heat from the water heater to the interior of the home.

As a result of homes becoming larger and the installation of inefficient hot



water distribution systems, some Americans are waiting minutes for hot water, or wasting a lot of money in their utility bills, or both. This article evaluates the performance of typical hot water distribution systems, and of various alternative systems and options that are designed to improve the performance of the typical system.

Modeling Systems

The Oak Ridge National Laboratory (ORNL) is developing a computer simulation model to evaluate the performance of residential hot water distribution systems in support of DOE's Building America program (see "Model Validation"). This model was also used in a California Energy Commission (CEC)-sponsored study to evaluate the performance of conventional and alternative distribution systems in California homes.

The objective of the California project was to simulate the energy and water performance, establish cost-benefit factors, and identify barriers to the

use of various domestic hot water distribution systems in typical new and existing California homes. (The effort to identify barriers to the use of various efficient hot water distribution systems continues and will not be reported on in this article.)

Comparing Hot Water Delivery Systems

The benefits of the various alternative distribution systems and options were based on an analysis of expected utility cost savings. Ten California cities (San Diego, San Bernardino, Fresno, Gilroy, San Jose, Sacramento, Davis, Fairfield, Tracy, and Stockton) were identified to represent the climatic and utility cost variations within the state. Utility costs in Tracy represented the average for these ten cities, and these costs were used in the analysis. They were \$0.11589 /kWh for electric, \$0.68263/therm for gas, and \$0.85/100 ft³ (base rate) for water.

Two use patterns were investigated. The first pattern assumed that each indi-

vidual draw was a cold start—that is, that the water in the pipe had reached the ambient temperature surrounding the pipe before each use. An all-cold-start use pattern represents the worst case for potential water and energy waste. The second pattern assumed was a clustered use, with individual draws clustered in the early morning and late afternoon and evening. When times between individual draws are short, the water standing in the hot water piping doesn't cool off completely. The clustered use represents the best case for water and energy waste.

The energy and water waste for the worst case was 2 to 5 times higher than it was for the best case. Real-world consumption patterns fall somewhere between these two extremes, but they are probably closer to the cluster use pattern than the cold-start pattern. However, the size of this spread shows that we need to learn more about actual hot water consumption patterns if we want to make distribution systems as efficient as possible.

The study examined new construction and existing housing typical of California. The new construction consisted of five examples. These ranged from a four-bedroom, two-and-a-half-bath, 3,080 ft², single-family, detached home to a one-bedroom, one-bath, 580 ft² apartment. The existing housing consisted of two examples. One was a three-bedroom, two-bath, 1,100 ft², single-family home; the other was a four-bedroom, two-and-

a-half-bath, 1,960 ft², single-family home.

All of the new and existing hot water systems had the following characteristics:

- The water heater was located in the garage.
- The house layout did not place the consuming devices near the water heater.

• Change piping materials used in the conventional system (see Figure 1), holding everything else constant.

- Move water heater to a more central location, shortening the length of the distribution system. Evaluate the impact for each of the piping materials.
- Add insulation to the various

piping materials in standard system configurations.

The following scenarios were evaluated for the impact of using selected alternative distribution systems in new construction:

- Install demand-actuated recirculating pump and controls in a conventional system for larger residences.
- Replace the conventional (trunk-and-branch) system with a continuous recirculation system (see Figure 2), for larger residences.
- Replace the conventional (trunk and branch) system with a parallel (radial) pipe/manifold system (see Figure 3), for all residences.

The following scenarios were evaluated for the impact of using alternative systems as upgrades or replacement options for existing housing:

- Upgrade an existing conventional system with a demand-actuated recirculating pump and controls.
- Replace an existing conventional system in kind. Evaluate the impact of the various alternative pipe materials and the addition of insulation.
- Replace an existing conventional

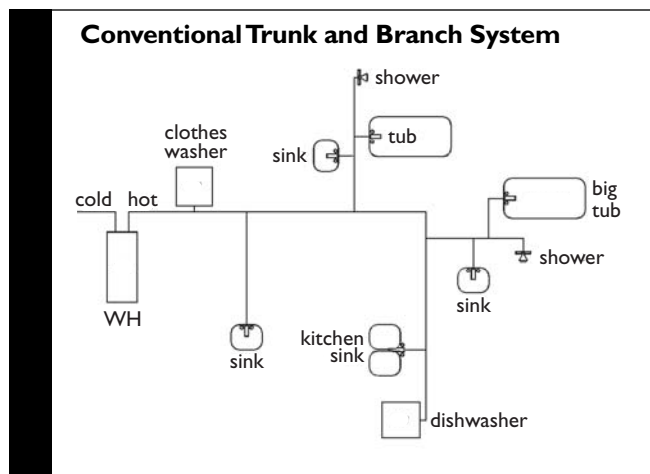


Figure 1. Shortening the distance between the water heater and fixture, keeping pipe sizes as small as possible, using CPVC pipe, and locating the system within the structure or in the attic insulation can markedly improve conventional hot water distribution system performance.

This resulted in significantly longer runs of distribution piping than is optimal.

- Pipes were located per standard California practice for the type of residence in question.

The following distribution system scenarios were evaluated for their impact on (1) energy and water waste, (2) the waiting time for hot water to arrive, and (3) the installation cost compared to a conventional distribution system:

Model Validation

Researchers and agencies interested in reducing distribution system waste have proposed that the model results described in this article be validated using appropriate field or experimental data. These studies have not yet been funded. The ORNL model does yield results that are generally consistent with the results of other models and generally with the experiences of those working in the field.

One particular area of uncertainty is the determination of appropriate soil conductivity factors for hot water distribution pipes below the slab. Using ASHRAE's values yields energy losses and waiting times well below the anecdotal information provided by some of those working in the field. Another area of concern is the water use pattern to be used in the model. We looked at an all cold starts pattern, where the water in the

pipe returns to ambient conditions before each draw and a clustered-use pattern, where the draws are clustered in the early morning and late afternoon and evening. Our study shows that the pattern and amount of water use significantly impact the analysis of system performance. A more thorough evaluation of actual water use patterns has been proposed but has not yet been funded.

system with a parallel pipe/manifold system.

Delivery System Performance

The energy and water waste performance among the scenarios for new construction was fairly consistent for all single-family detached houses studied. Assuming a clustered-use consumption pattern, a conventional system with demand recirculation had the lowest water and energy waste (see Table 1). A centralized water heater (located adjacent to the kitchen) was next lowest in terms of energy and water waste, but one of the houses we studied still had an unacceptable waiting time for the arrival of hot water (greater than 30 seconds). The parallel pipe/manifold system with the clustered-use pattern performed less well than conventional systems. However, this system performed significantly better than the conventional systems with the more wasteful, cold-start use pattern. All of these systems had acceptable typical waiting times (less than 30 seconds), although the conventional system had much longer maximum waiting times (58–98 seconds).

Among the conventional trunk-and-branch systems studied, chlorinated polyvinyl chloride (CPVC) pipe outperformed copper pipe in all scenarios in reducing waste.

Locating the conventional-system pipes in the attic, buried within the loose-fill insulation, was less wasteful than locating the pipes in the crawlspace. Locating the pipes below the floor slab was the most wasteful for conventional systems when the cold-start use pattern is modeled, but not when the clustered-use pattern is modeled. Continuous-recirculation systems had, by-and-large, a short waiting time and low water waste, but

they also had by far the highest energy costs. Locating continuous-recirculation systems in the attic within the attic insulation was far less wasteful than locating them below the floor slab.

For existing housing, a more limited set of scenarios was studied. One

replacement system. The energy savings of this system depended on the water use pattern assumed. For the clustered-use pattern the energy consumption of the parallel pipe system was about 20% higher than a conventional system; it was about 50% lower for all cold starts when compared to a conventional system.

The other existing-housing scenario involved homes with an unacceptable waiting time for hot water. Installing a demand recirculation pump and controls reduced the waiting time to the lowest level of the scenarios studied and reduced energy and water waste by about 50% compared with the existing conventional copper system, assuming a clustered-use consumption pattern.

Continuous-recirculation systems require the installation of a return pipe, which is not practical for most existing situations. The water delivery time is comparable for continuous- and demand-recirculation systems, if one doesn't count the delay from pushing the button on the demand system. Including this delay would add 5 to 20 seconds to the demand systems, depending on the length of the system, the size of the pump installed, and other factors.

The Bottom Line

For households that heat water electrically, the cost difference between the most wasteful—continuous recirculation—and least wasteful—demand recirculation—systems is about \$320 per year (see Table 1). With gas water heaters, the potential saving is more modest—about \$100 per year. The water saving between the most and least wasteful systems is about 1,640 gallons per year. Some of the more efficient options also cost less, making them real winners. Compared to the standard-practice conventional system with copper pipe located below the floor slab, the following systems cost

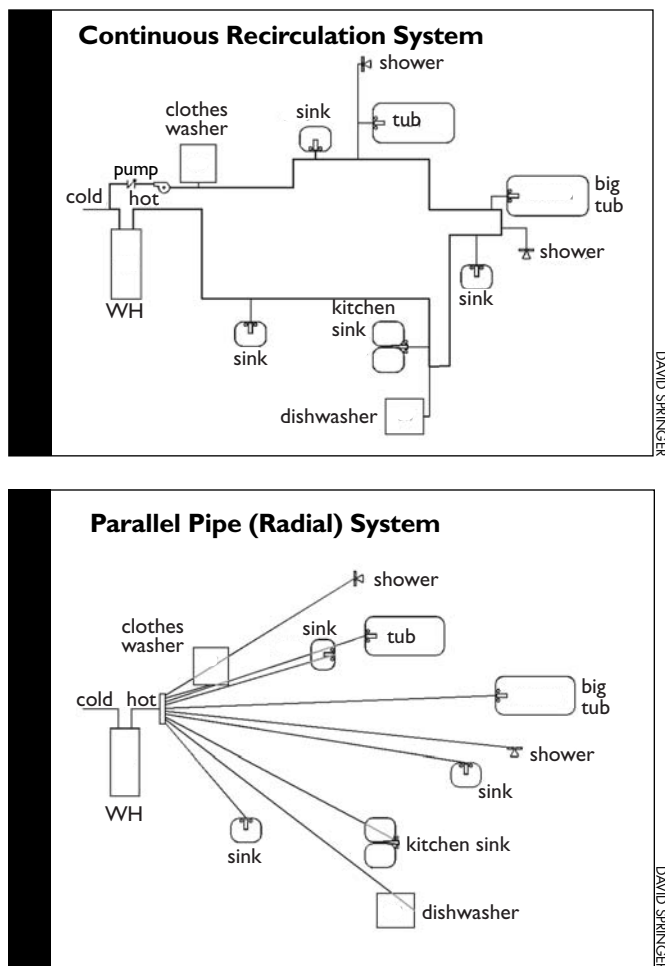


Figure 2. (top) Continuous-recirculation systems are typically installed in large houses with long distances for hot water distribution. While they save water and shorten waiting times, they consume a lot of energy. Making the improvements suggested for a conventional system and adding controls to the pumps can help improve the system's energy performance. **Figure 3.** (bottom) Parallel pipe systems are performing well in a small number of homes. An individual pipe serves each end use from a manifold usually located near the water heater. The overall performance of this system is significantly impacted by the water-use pattern in the home. Small diameter PEX tubing (as low as 3/8-inch) reduces water volume and heat transfer through the tubing, which enables the system to outperform conventional systems under some circumstances.

scenario involved replacing a conventional system. In this case a parallel pipe/manifold system with cross-linked polyethylene (PEX) tubing both reduced the waiting time for hot water and reduced the installation cost of the

Table 1. Hot Water Distribution Systems and Parameters

Evaluated for a Three-Bedroom, Two-Bath, 2,010 ft² Home, Using a Clustered Hot Water Use Consumption Pattern

System/Option Evaluated	Construction Cost	Wait Time for HW (Sec.)		Annual Water Waste (Gallons)	Annual Energy Waste (\$)	
		Typical	Max.		Electric@ \$0.116/kWh	Gas @ \$0.683/Therm
Conventional, attic, copper—central water heater	\$1,450.00	5	40	1421	35.04	10.92
Conventional, attic, CPVC—central water heater	\$1,087.00	5	39	1406	33.00	10.32
Conventional, attic, copper	\$1,271.00	5	98	2357	58.32	18.24
Conventional, attic, copper—insulated	\$1,552.00	5	98	2357	58.32	18.24
Conventional, attic, CPVC	\$ 866.00	5	95	2287	54.24	16.92
Conventional, attic, CPVC—insulated	\$1,147.00	5	94	2280	54.12	16.92
Conventional, slab, copper	\$1,556.00	4	70	1697	45.12	14.04
Conventional, slab, copper—insulated	\$1,838.00	4	60	1560	40.20	12.60
Conventional, slab, CPVC	\$1,086.00	4	58	1549	36.84	11.52
Conventional, slab, CPVC—insulated	\$1,368.00	4	58	1542	36.60	11.40
Demand recirculation, attic, copper	\$1,880.00	5	9	940	26.16	8.16
Demand recirculation, slab, copper	\$2,447.00	4	8	814	23.64	7.44
Demand recirculation, attic, CPVC	\$1,475.00	5	9	941	23.64	7.44
Demand recirculation, slab, CPVC	\$1,978.00	4	8	844	20.88	6.48
Parallel pipe, attic, PEX	\$1,226.00	11	36	2344	59.64	18.60
Parallel pipe, slab, PEX	\$1,443.00	11	38	2455	64.08	20.04
Continuous recirculation, attic, copper—insulated*	\$2,559.00	5	9	940	146.28	45.72
Continuous recirculation, slab, copper—insulated*	\$2,861.00	4	8	814	340.68	106.44
Continuous recirculation, attic, CPVC—insulated*	\$1,965.00	5	9	941	144.96	45.24
Continuous recirculation, slab, CPVC—insulated*	\$2,185.00	4	8	844	315.00	98.40

*Pumping power consumption for the continuous recirculation systems adds \$87.60 to both gas and electric totals shown above.

less to build:

- systems made with CPVC and PEX piping;
- systems located in the attic;
- systems with a centrally located water heater;
- demand recirculation systems if CPVC pipe is used; and
- parallel pipe systems using PEX pipe.

What Should Builders Do?

Although this study modeled hot water delivery systems, and, as we all know, real life often confounds the models, we can make some solid recommendations about moving toward more efficient and less costly delivery systems.

- Consolidate hot water-consuming devices in the same areas, to take advantage of clustered uses of hot water.
- Centrally locate water heaters to minimize runs of distribution piping.
- Locate plumbing in the attic (buried in the insulation) for single story homes without basements and in the space between floors for multistory homes to minimize energy loss through the pipes.

• Do not oversize hot water piping. Use code permitted minimums. Bigger isn't better.

• Consider a demand recirculation (one-pipe) system in lieu of a continuous-recirculation (two-pipe) system if waiting time is an issue.

• Consider CPVC or PEX plastic piping in lieu of copper when appropriate quality and durability can be demonstrated.

What Should Home Buyers Do?

An individual house will perform differently depending on the hot water use patterns of the household.

However, we recommend that people who are shopping for a house do the following things before they buy:

- Time how long it takes for hot water to arrive at the most important fixtures, such as the master bath shower. Do this several hours after the fixture was last used to simulate the waiting period for the first use of the day.
- Note the distance between the water heater and the furthest hot water-consuming fixture. Long systems have

more waste and delay.

• Find out what the pipes are made of, whether they are insulated, and where the hot water system is located. Uninsulated copper pipe below the floor slab can have more waste and delay than other systems.

• Ask what controls are provided for any recirculation systems that have been installed. Uncontrolled systems have very high energy costs.

Researchers and agencies interested in reducing distribution system waste will continue to gather data on how builders are building hot water distribution systems, and how people are using those systems. Given the reality of increasing energy needs, instability in world markets, and concern about global warming, saving energy with more efficient hot water systems is not an option, but an imperative.



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